

**CRASH
CART**



CRASH CART

- ▶ A **crash cart** is a set of trays on wheels used in hospitals for transportation and dispensing of emergency medication/equipment at site of surgical emergency for life support protocols to potentially save someone's life.
- ▶ Should be kept in easily accessible position- central to patient care areas



Crash cart - nut shell



Emergency equipment



Medications



Intubation Supplies



IV Start Supplies



Procedure Trays

REAR VIEW

Back of Cart

Backboard

Clipboard

Procedures Sheet

Crash Cart Check Off List

Pharmacy Charge Sheet



FRONT VIEW

Bottom Basket

- Large Gloves
- Defibrillator Pads
- Multi-function Electrodes
- Surgical Cone Mask



Top Basket

- BVM with Large & Medium Masks

Right Side of Cart

- Sharps Container



SIDE VIEW

► Left side of Cart

Tank & Gauge
Crash Cart Inventory List
Anaphylaxis Treatment Guide



First Drawer: Medications

- Amiodarone - antiarrhythmic
- Bretylium tosylate -ventricular antiarrhythmic
- Nitroglycerine -vasodilator
- Atropine –parasympatholytic
- Vasopressin
- Dopamine
- Epinephrine
- Anesthetics- ketamine , xylazin
- Lidocaine
- Antibiotics
- Naloxone -opiate antagonist

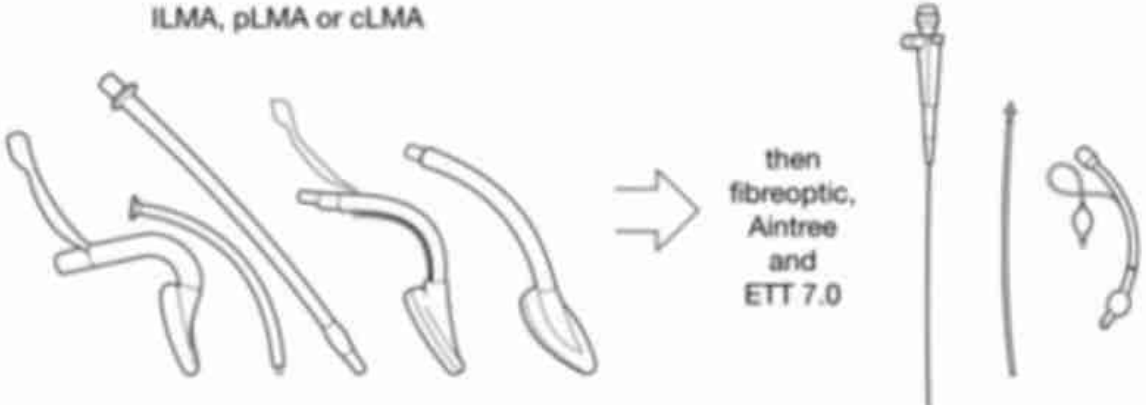
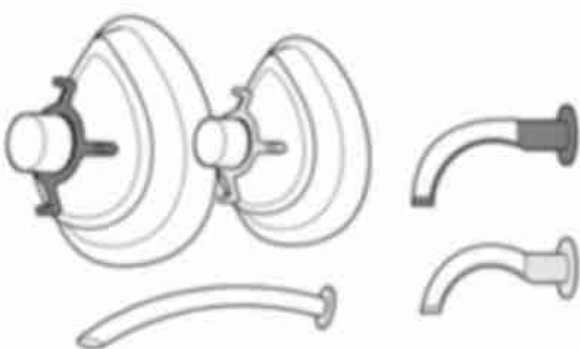
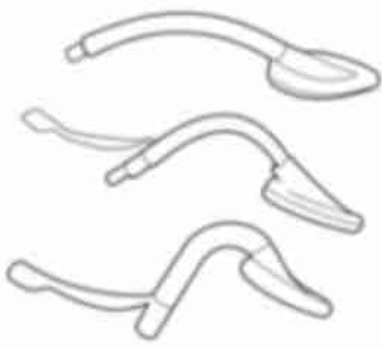




First Drawer: Medications -ctd.

- Sodium bicarbonate
- Magnesium chloride
- Sodium Chloride 0.9%
- Calcium Chloride
- Sodium Bicarbonate
- Sterile Water Inj. 20 ml vial
- Dextrose 50%
- Povidone-Iodine Swab stick
- Alcohol Swabs

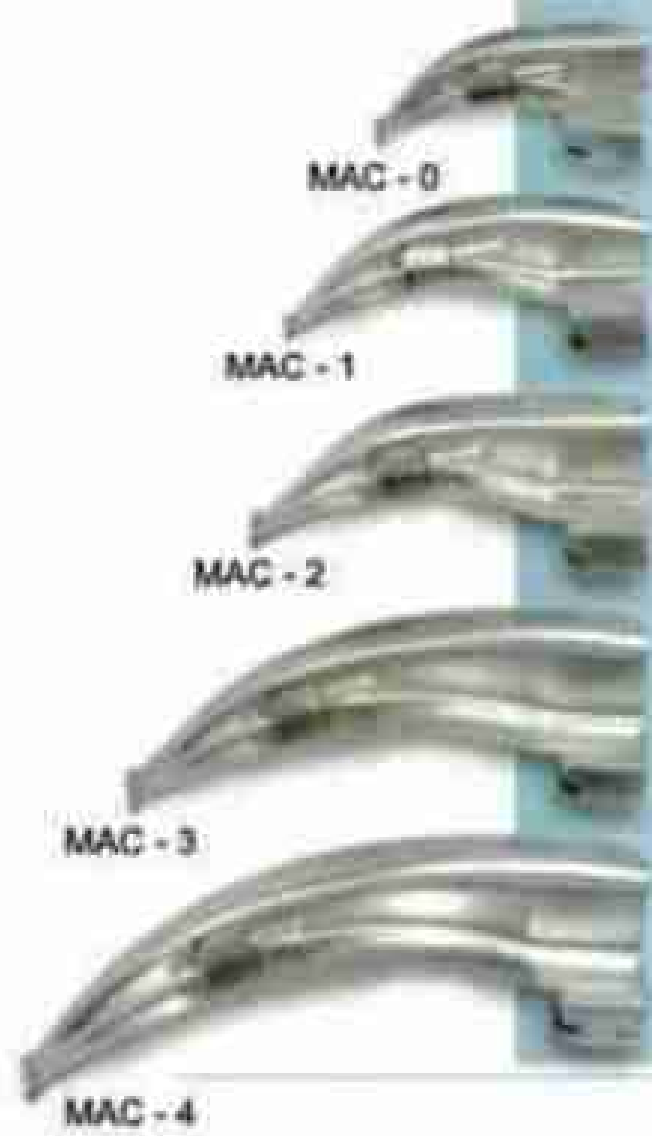


Plan A: Initial intubation strategy Elective intubation max 4 Rapid sequence induction max 3	Optimum position  Bougie  Alternative laryngoscope  C-MAC 
Plan B: Secondary intubation strategy Not in rapid sequence RSI	ILMA, pLMA or cLMA  then fiberoptic, Aintree and ETT 7.0
Plan C: Oxygenation and ventilation Wake patient up Consider sugammadex	Face mask, oro- or nasopharyngeal  cLMA, pLMA or ILMA 
Plan D: Can't intubate, can't ventilate ICV 	Melker  Quicktrach  Manujet and jet ventilation catheter  Surgical airway 

Drawer 2: Intubation Supplies

- Endotracheal tubes
- Laryngoscope
- Gauze
- Nasopharyngeal Airways
- Tongue Depressor





MAC-0

MAC-1

MAC-2

MAC-3

MAC-4



CONVENTIONAL ENGLISH
MILLER BLADES



MIL-0

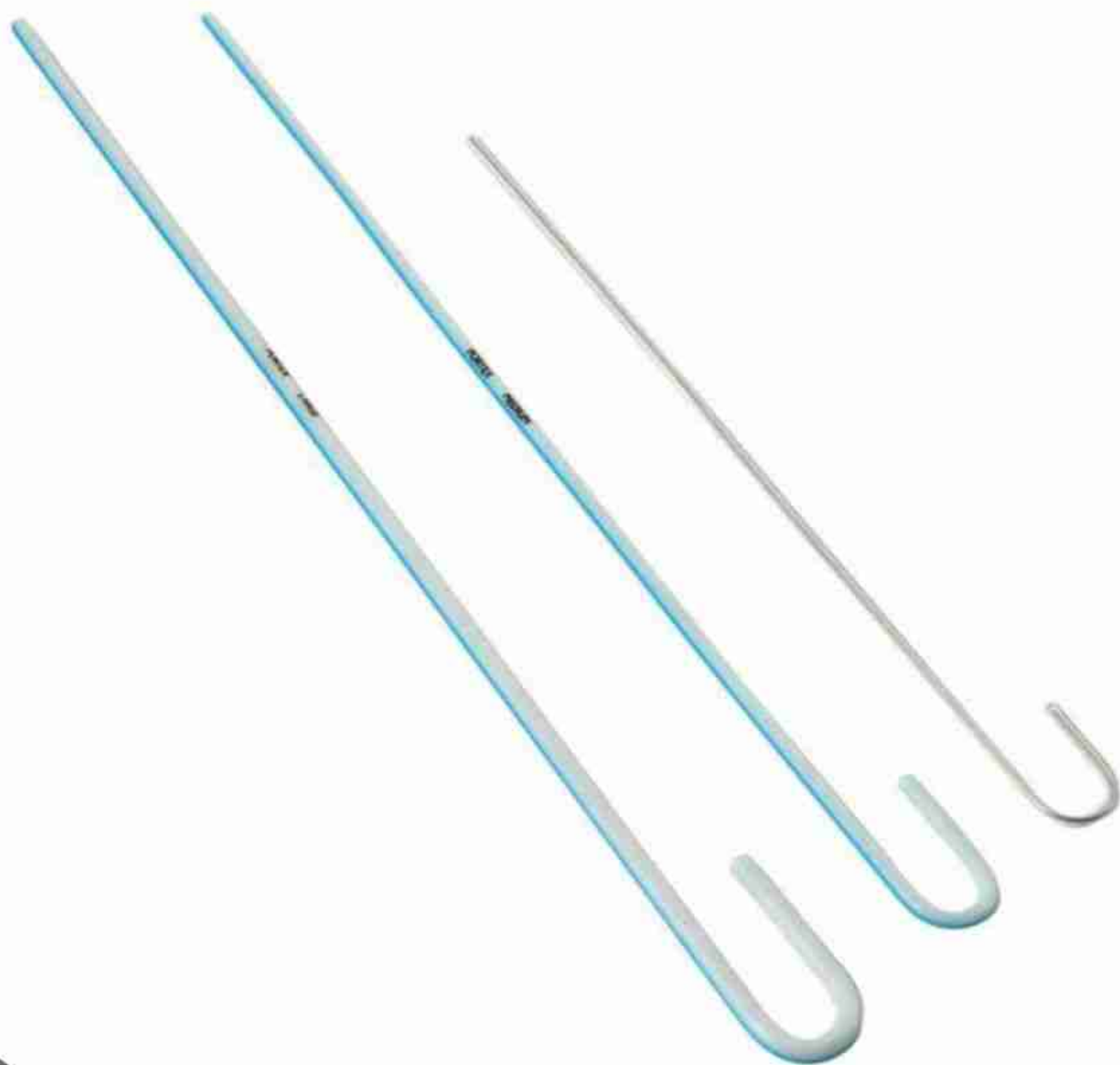
MIL-1

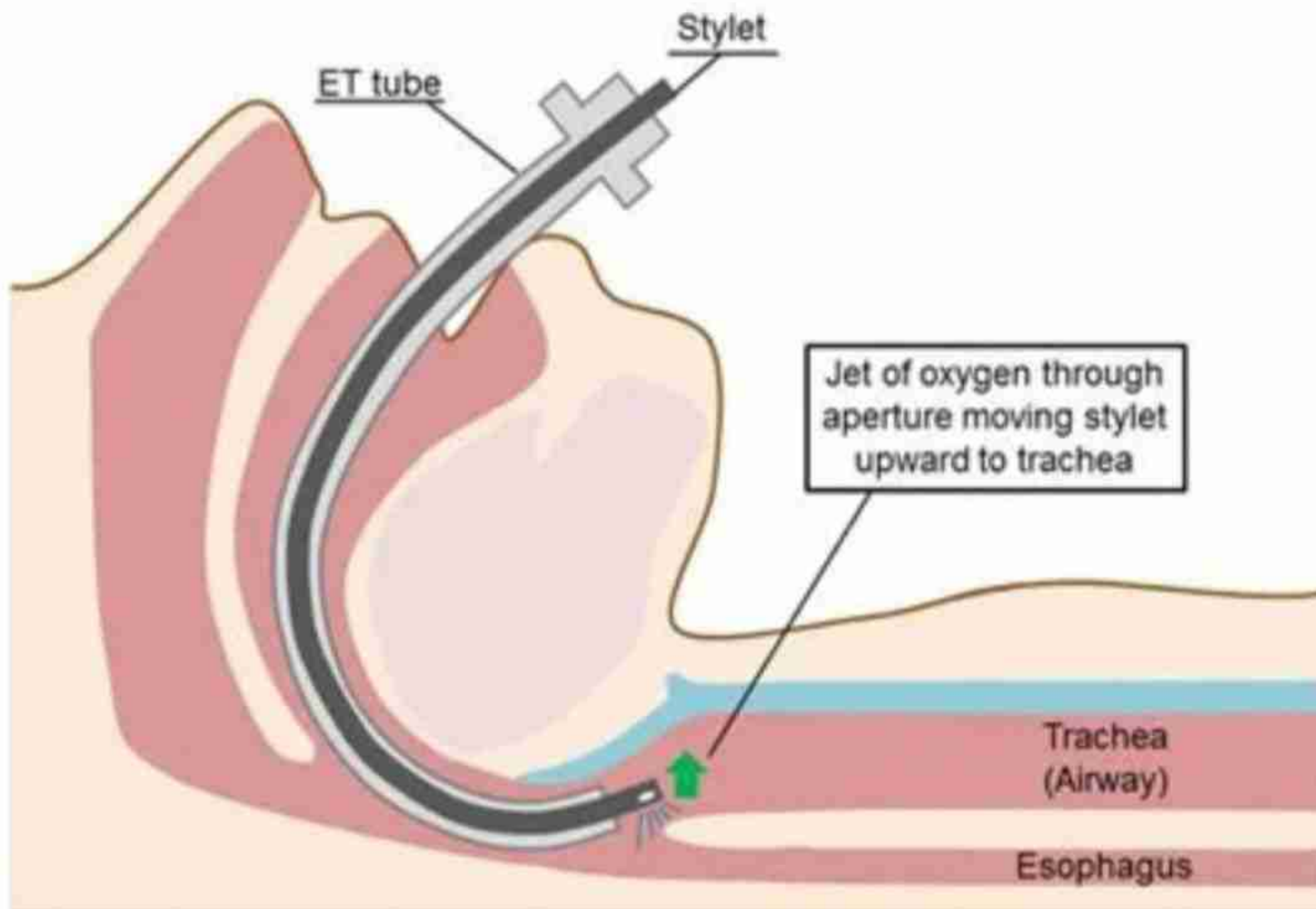
MIL-2

MIL-3

MIL-4

CONVENTIONAL ENGLISH
MACINTOSH BLADES





The term “stylet” is somewhat elastic. It can refer to rigid or malleable rods or wires used to adjust the curve of an endotracheal tube in order to make insertion easier.

15 mm Connector



Aperture Bars

Airway tube

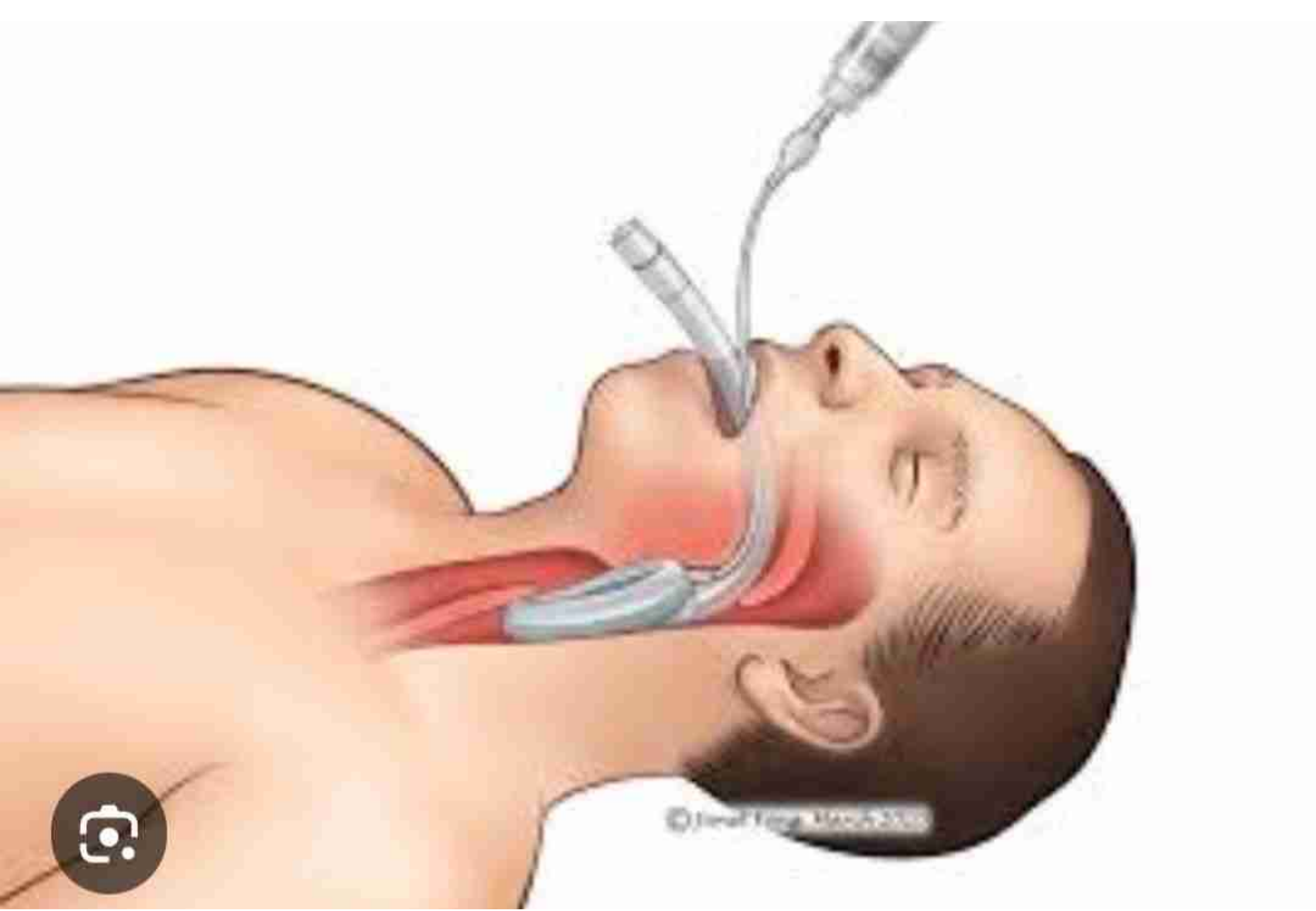
Cuff

Inflation line

Inflation
Pilot Balloon

Valve





Indications for the use of the LMA



- Situations involving a difficult mask (BVM) fit.
- May be used as a back-up device where endotracheal intubation is not successful.
- May be used as a "second-last-ditch" airway where a surgical airway is the only remaining option.

Contraindications of the LMA



- Greater than 14 to 16 weeks pregnant
- Patients with multiple or massive injury
- Massive thoracic injury
- Massive maxillofacial trauma
- Patients at risk of aspiration
- **NOTE:** Not all contraindications are absolute

Side-Effects of the LMA



- Throat soreness
- Dryness of the throat and/or mucosa
- Side effects due to improper placement vary based on the nature of the placement



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1

2

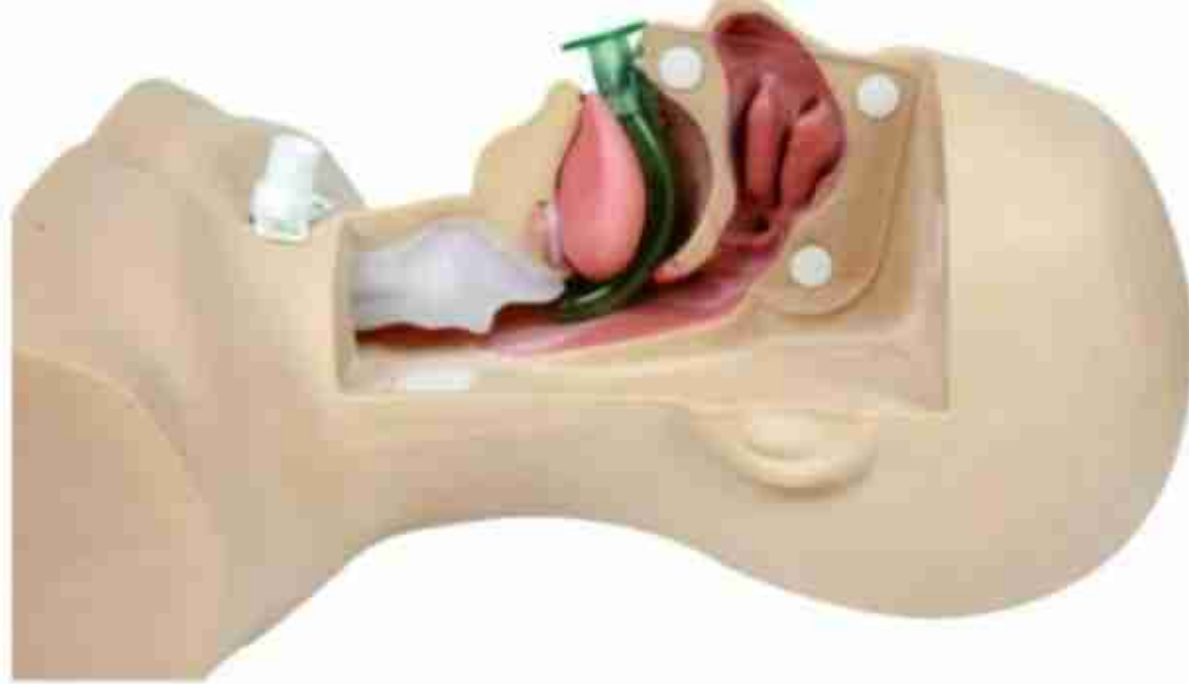
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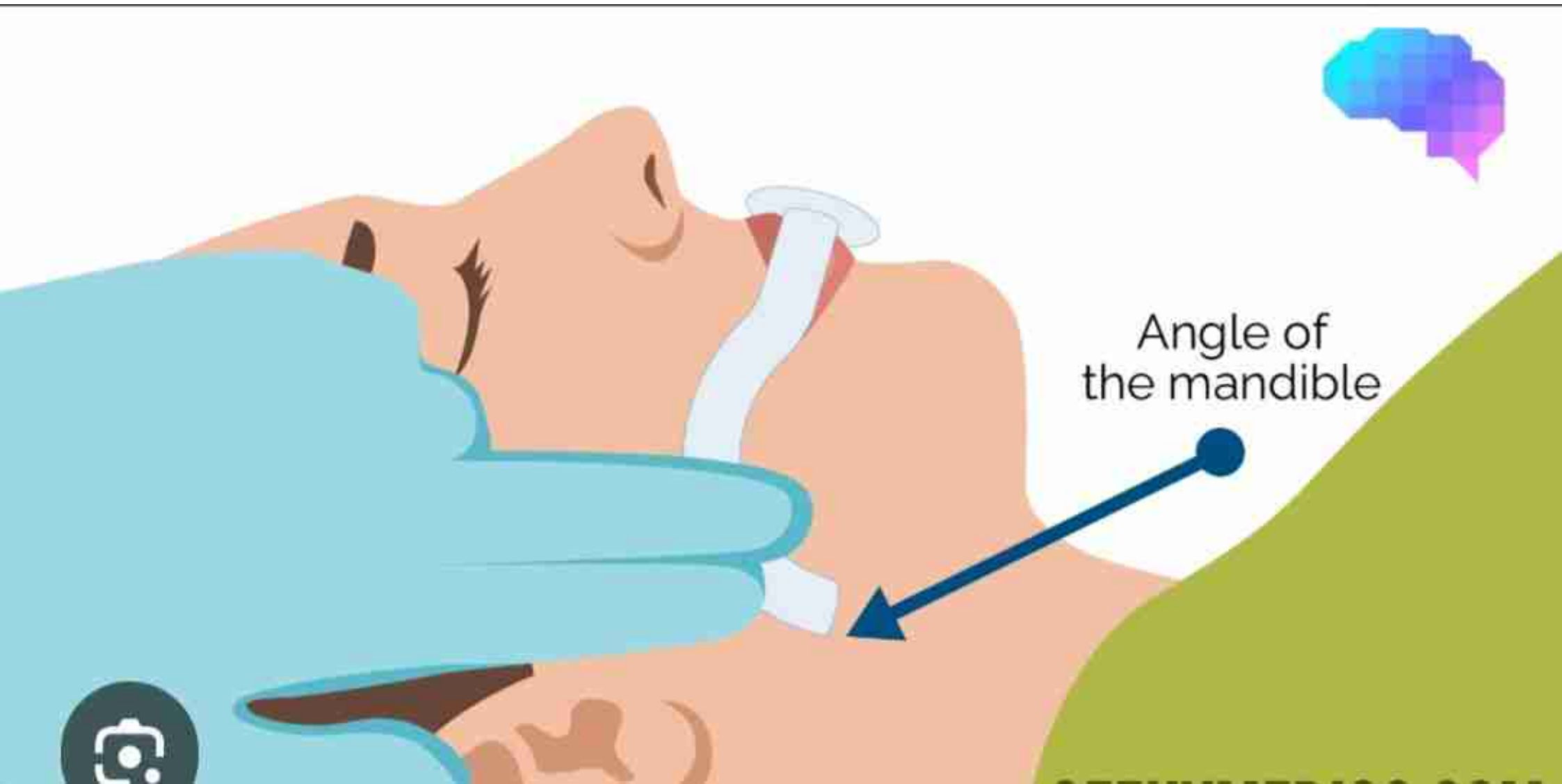
Oropharyngeal Airway

- Indications
 - OP Airways are indicated in unconscious patients without a gag reflex
 - An Oral Airway will prevent the tongue, which is the most common airway obstruction, from blocking the airway when properly inserted.





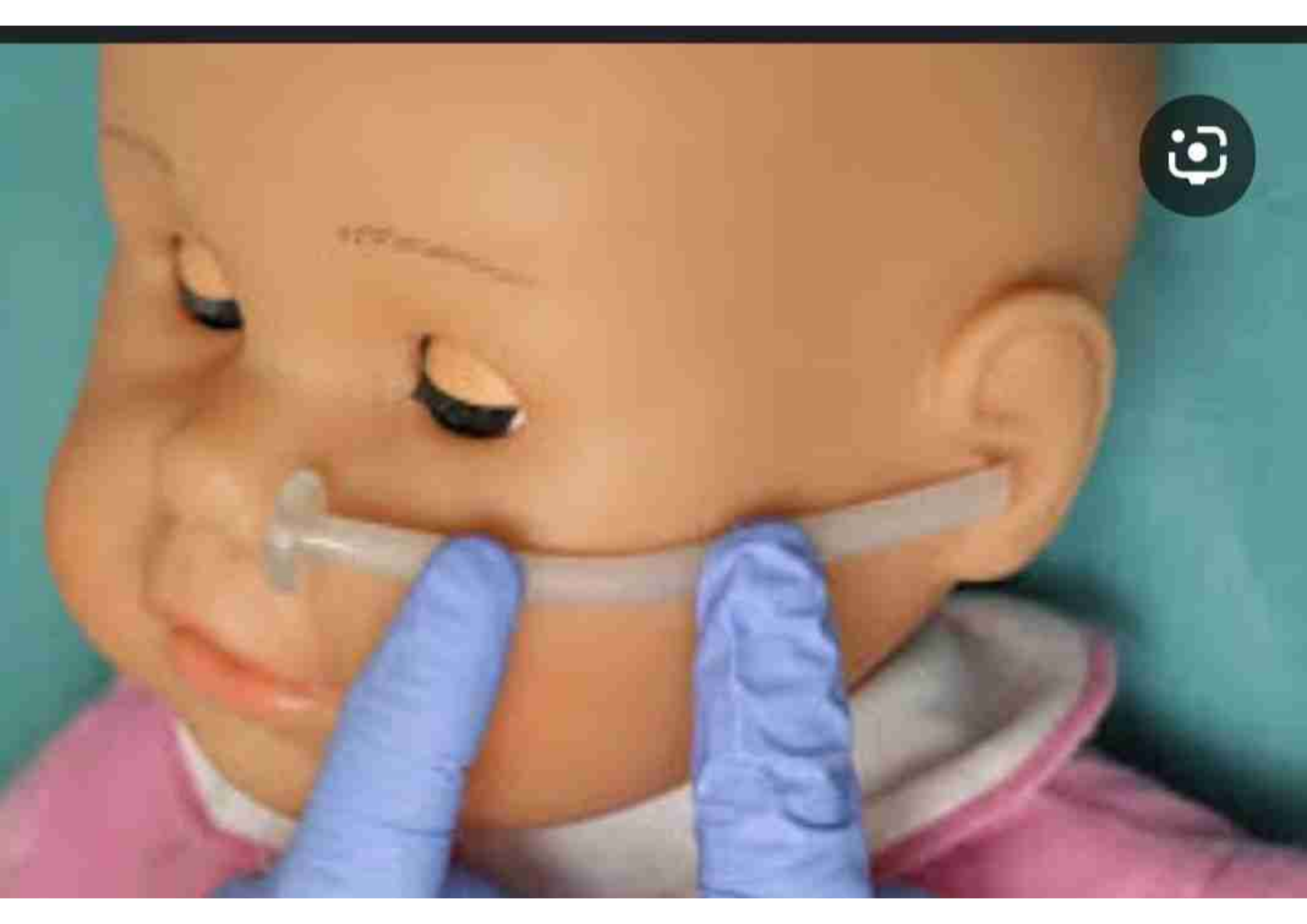
1. Select the appropriate size measuring from the corner of the mouth to the base of the earlobe or angle of the jaw
2. Open the mouth using the jaw thrust or cross-finger technique.
(consider c-spine injury)
3. Insert the airway with the tip pointed toward the hard plate without pushing the tongue posteriorly
4. Rotate 180 degrees when no resistance is felt and the airway reaches the hard palate. Or insert from the side of the mouth with the tip pointed toward the inside of the cheek.





Indications:

- NPA uses conscious, semiconscious, or unconscious patient.
- NPAs are also helpful when a patient's mouth is difficult to open or access, for example, in cases of trismus or angioedema.
- OPA is not possible.
- Gag reflex, massive trauma around the mouth.





Drawer 3: IV Start Supplies

- Syringes-3,6,12,20ml -drug administration
- Needles-14,16,18,20,22,25 gauge -drug administration
- IV Catheters-16,18,20 gauge - drug administration
- Intraosseous catheter or bone marrow needles-18,20 gauge
- Canula
- Scalp vein set
- Vacutainers

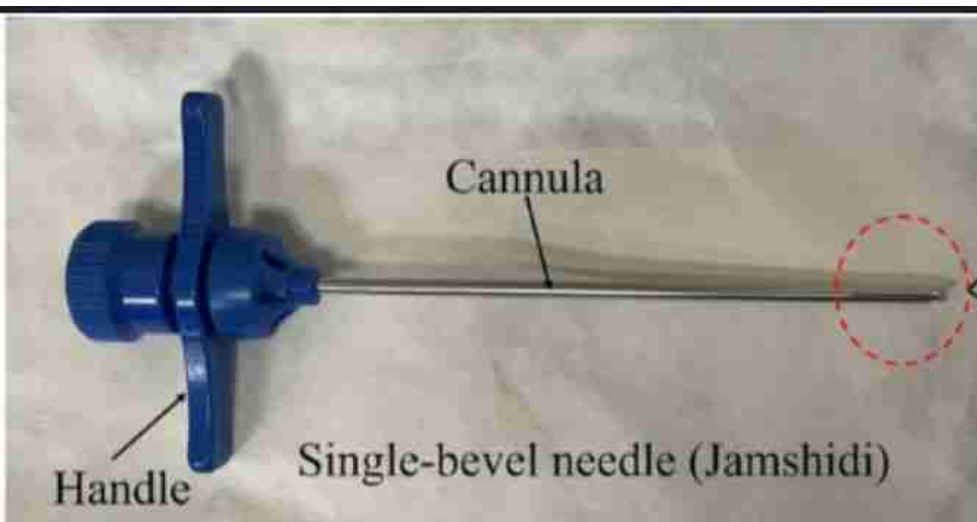




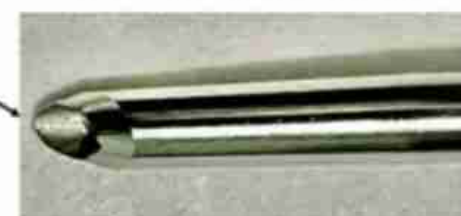
Drawer 4: IV Supplies and Tubing

- 30 ml or 35 ml Syringe
- Adhesive Tape 1"
- Catheter Tip Syringe
- Betadine solution

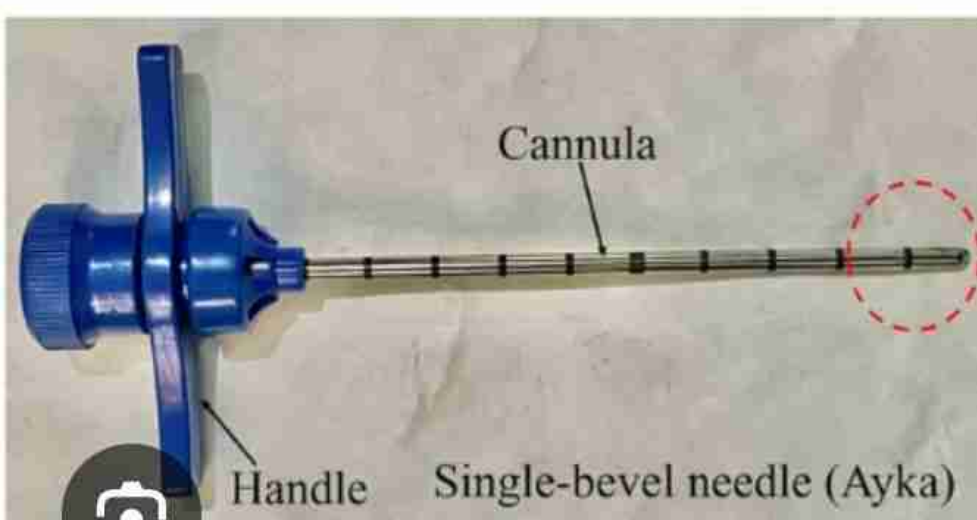
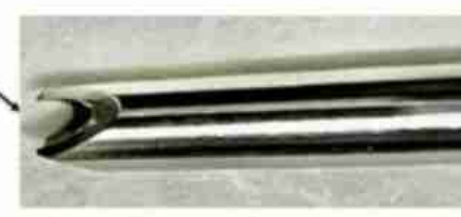
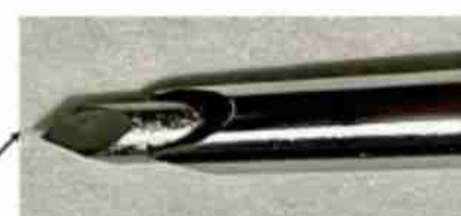




(a)



(b)



(c)





Cook IO Needle



Jamshidi IO Needle



Illinois Sternal Iliac Needle



Sur-Fast IO Screw Tip
Needle

What is an Intraosseous Access



- Needle inserted into bone
- “Non collapsible vein”
- Infuses into systemic circulation via bone marrow cavity
- Used for fluid/drug administration
- Able to aspirate marrow for bloods
- Equal predictable drug delivery and pharmacological effect

When is Intraosseous Indicated

- Difficult or failed IV access
- Life threatening or emergent situations
- Obese patients with limited vascular access
- Pre-hospital (extraction, moving vehicles)

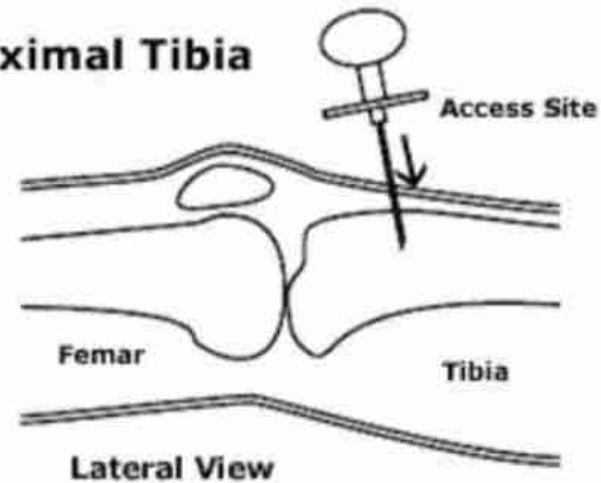
Common insertion sites for paediatric IO access



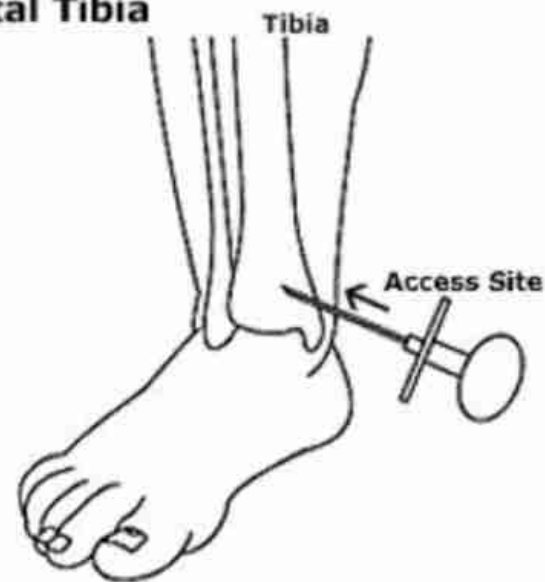
- Identify the appropriate site
 - Proximal tibia: Anteromedial surface, 2-3 cm below the tibial tuberosity
 - Distal tibia: Proximal to the medial malleolus
 - Distal femur: Midline, 2-3 cm above the external condyle



Proximal Tibia



Distal Tibia



Drawer 5: Procedure Trays

- ECG Electrodes
- Surgeon's Gloves- Sizes 6, 6½, 7, 7½, 8
- Sutures
- Needle Holder
- Sterile Field



- Stethoscope
- Cotton
- Tape
- Razores
- Sissorses
- Flash light
- Normal saline



Maintenance of Crash Cart

- ▶ It is important that the crash cart be checked regularly and **maintained** so that its contents are there when needed.
- ▶ **Expiration dates** on medications should be checked on the first day of the month
- ▶ Expired medications should be promptly **removed** and **replaced**
- ▶ The defibrillation pads on the defibrillator should be checked for **expiration** date
- ▶ The **battery charge** on the monitor should be checked and documented

CRASH CART POLICY





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	Title	DEPTNAME	Document Type
1	CRASH CART POLICY	National Heart Center	Policy

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	Sultanate of Oman Ministry of Health The Royal Hospital Department of National Heart Center
Code: NHC-GNL-POL-1-Vers.1	
Effective Date: 23/08/2022	
Target Review Date: 23/08/2024	
Title: Crash Cart Policy	

Royal Hospital

Crash Cart Policy- Contents, Use and Quality Assurance

A.	Policy Statement:	1. To inform the Royal Hospital employees what resources and equipments are available for Code Blue/Medical Emergencies (Cardiac Arrest). 2. To standardize all the crash carts with same equipments and emergency medical supplies and provide quality assurance for managing Code Blues/Medical Emergencies
B.	Definitions:	1. Code Blue/Medical Emergency is defined as any medical emergency i.e., cardiac arrest, respiratory arrest, neurological incident (seizure), or any time a response emergency team is required. 2. Crash Cart is a trolley designed for storing lifesaving equipments and drugs. It is allocated in an accessible area for all health care providers in order to provide the patients with the necessary emergency care in life threatening events such as treatment room and resuscitation room.
C.	Purpose :	1. To have a Crash cart available in all department that treats patients who have the potential to have a sudden deterioration in their condition. 2. To establish standard practice for maintaining and utilizing the crash cart. 3. To have a specific crash cart type that facilitates staff familiarity with equipment. 4. To save the valuable time at the time of emergency.
D.	Scope:	1. To be used by all health care providers.
E.	Policy:	1. Crash cart should be available in all clinical areas and kept always in sight that is easily accessible.

		2. Crash cart to be maintained and supplied in accordance with the crash cart content list. 3. The crash cart must be checked <u>once a day by two staff nurses</u> for the content availability, expiration and functioning. 4. The crash card must be locked/ sealed with label indicating that is checked and ready to use. 5. Additional supplies and / or equipment required may not be added to the crash cart. It should be maintained and stored separately. 6. Restocking cart contents after use to be done immediately by the same the shift assigned nurse. 7. Nurse In Charges are responsible to monitor and ensure that the crash cart is complete with all items as per checklist and is kept ready for use at all times. 8. Pharmacist should check emergency cart medications for proper storage, stock levels, rotation of near expiry medications and maintaining of Inspection log book every 2 months. 9. All equipment's present on the trolley should be according to the standard Royal Hospital crash cart content checklist.
F.	<i>Procedure</i> 1. Checking and restocking	1. Two nurses will be assigned in morning shift to be responsible for cleaning the cart, inspecting and replacing emergency drugs as well as checking the proper functioning of the defibrillator, cardiac monitor, autoclaved items, oxygen cylinder, suction machine and all other items of the crash cart. 2. Drawers of crash carts should be clearly labeled to identify contents under the following categories e.g. 2.1 Drawer- 1: Emergency Medication 2.2 Drawer -2: Air way & Breathing Items 2.3 Drawer -3: Circulation / I.V Supplies 2.4 Drawer- 4: I.V Fluids & PPE items.

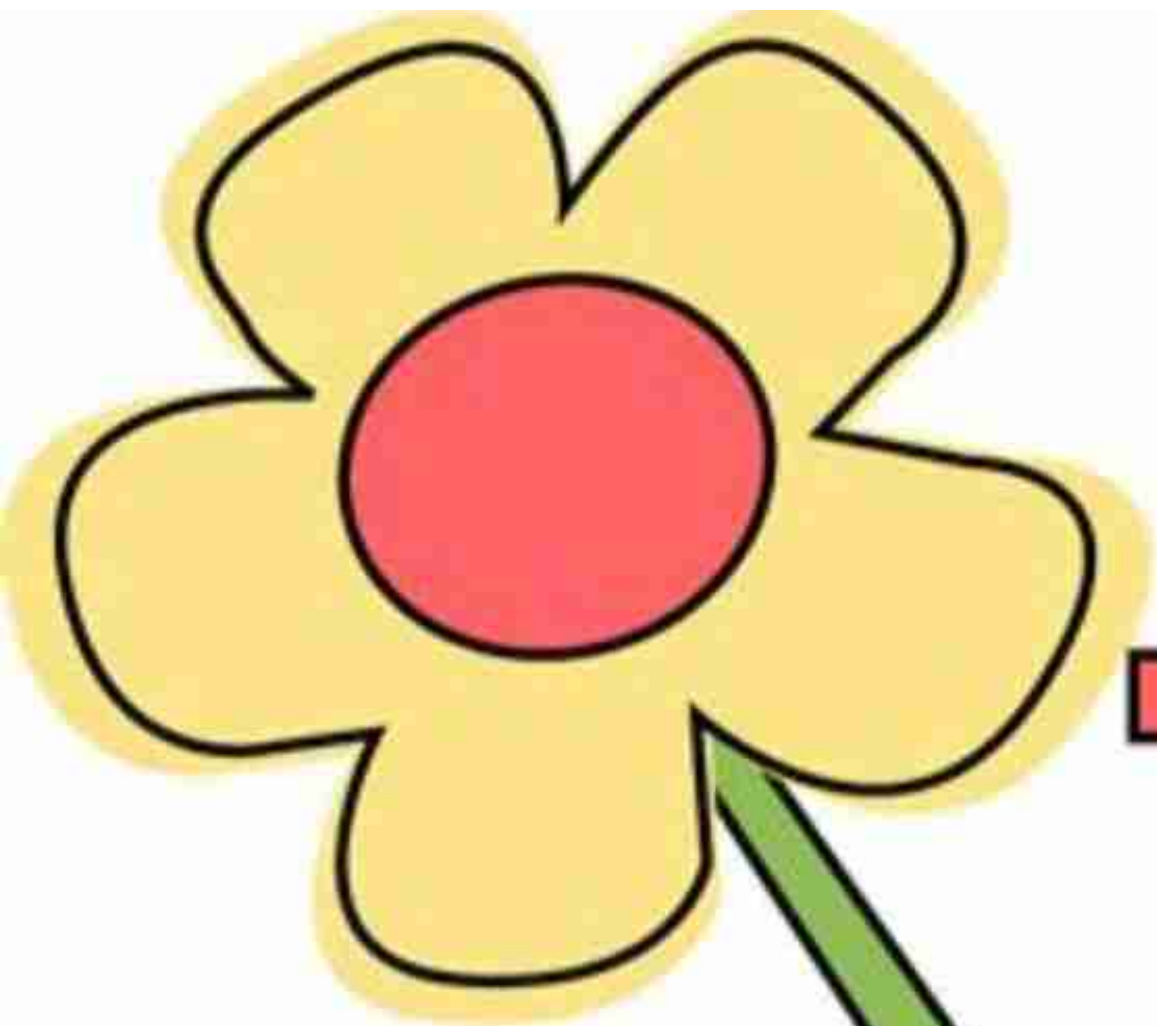
3. The drugs are present in accordance with standards in terms of location, quantity and dates of expiry.
4. An expiration label should be placed on the drug shelf/drawer to insure that no expired drugs remain in the shelf/drawer. Medications that are to expire within 30 days should be replaced.
5. The defibrillator is charged and tested for its performance on both battery and electrical current once in 24 hrs.
6. The defibrillator will remain plugged into an electrical outlet at all times, except during battery testing and according to the manufacture instruction.
7. Laryngoscopes with its batteries & light should be checked daily for its function. The blades should be clean with disinfected solution after each check.
8. Oxygen cylinders (2 nos.) should be kept and replaced when cylinder oxygen capacity is < 500 psi. One cylinder to be kept
9. Full as spare at all times.
10. After conducting a check the staff is responsible to document it on the crash cart checklist and sign. The checklist and test load strips should be maintained for a period of 7 days after use.
11. Crash cart and cart wheels to be kept locked if not in use.
12. Other emergency equipment such as IV and / or Intubation supplies, sterile items should be checked by assigned staff to ensure integrity of stock and equipment. Items with near expiry dates to be replaced within one month.
13. After the arrest, the crash cart must be restocked immediately within the same shift then sealed or locked.
14. The Biomedical department to be contacted immediately when a defibrillator or any other equipment problem is detected.

G. 1.2 Training

1. All nurses have attended Emergency Drugs & Crash cart training followed up by regular updates related to the use of crash trolley

		equipment and drugs. 2. All nurses should be familiar with the use of the contents and location of all medications and equipment in the crash cart. 3. Institutions are responsible to organize drills minimum twice per year on use of crash cart so as to keep the skills and competencies of doctors and nurses updated. 4. Nursing staff are competent in the administration of emergency drugs in terms of dosage, action, compatibility, side effects and contraindications.
H.	1.3 Quality Assurance (Q/A)	1. Cardiopulmonary Committee members conduct frequent audit to ensure the sustainability of having up to standards quality and maintenance. 2. Provide and action plan in regards to Crash cart quality assurance management. The plan addresses equipment, supplies, medications, inventory management, emergency protocols, training and competency of staff, emergency drills and/or simulation exercises, and assignment of responsibility for continued oversights of the process. 3. The action plan should be continuously reassessed, and revised to ensure that the crash carts and staff are ready for life-threatening medical emergencies.
I.	1.4 Maintenance	Crash cart equipments are scheduled for regular biomedical checks as per a preplanned Periodic Preventive Maintenance (PPM) schedule and records to be maintained. The following is a maintenance routine that should be completed at least monthly: • Expiration dates on medications should be checked on the first day of the month • Expired medications should be promptly removed and replaced • The defibrillation pads on the AED or the defibrillator should be checked for expiration date.

		The battery charge on the monitor and/or AED should be checked and documented.
J.	<i>Outcome</i>	- Crash cart is well equipped as per standard checklist and is readily available for use at all times. - All Doctors, Nurses and Pharmacist working in the Hospitals have been oriented and trained to use the crash cart and its content.
K.	<i>References</i>	https://nursingmanthra.com/2017/09/05/crash-cart-arrangement-in-hospital-settings/ https://www.jointcommission.org/resources/news-and-multimedia/newsletters/newsletters/quick-safety/quick-safety-issue-32-crashcart-preparedness/crashcart-preparedness/



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